

TrES-3b

R-2115

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130314 · created 2026-08-11T00:39:17+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456365.93675 +/- 0.00098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.011
Transit depth (Rp/Rs)^2	3.54 +/- 0.4 [%]
Orbital Inclination (inc)	83.25 +/- 0.53
Airmass coefficient 1 (a1)	1.201 +/- 0.014
Airmass coefficient 2 (a2)	-0.1289 +/- 0.0081
Scatter in the residuals of the lightcurve fit is	1.17 %
Best Comparison Star	None
Optimal Aperture	3.58
Optimal Annulus	10.17
Transit Duration (day)	0.0681 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.188 $\pm$ 0.011 vs 0.16309 (deviation 1.96 σ)
Duration fit vs published	0.0681 d vs 0.0567 d (deviation 3.0 σ)
Mid-time O-C	0.3 min
Depth SNR	14.8
Residual scatter	1.17 %

Light curve

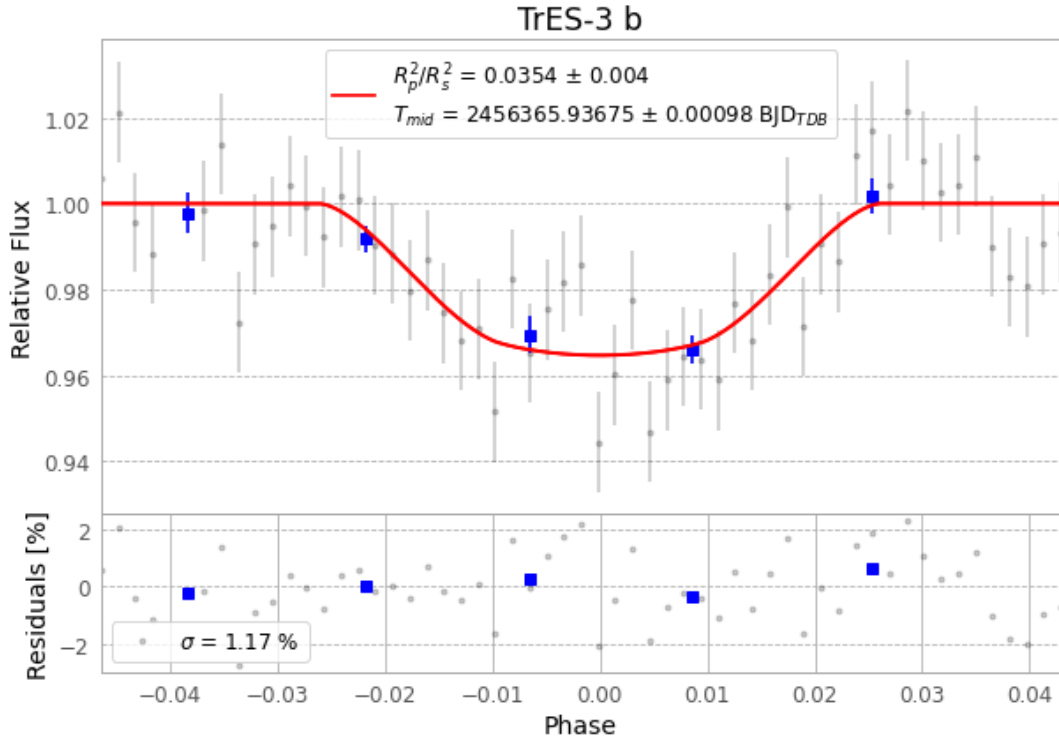


Figure 1 — FinalLightCurve_TrES-3 b_2013-03-14.png (SHA-256 f2f52cd5ab4e8398...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [298, 181], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a108dd1e61f92911...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

57 FITS frames (37 MB), TRES-3130314090012.FITS → TRES-3130314114812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130314.log (ff35d2d8cf79...), FinalLightCurve_TrES-3 b_2013-03-14.png (f2f52cd5ab4e...), FinalLightCurve_TrES-3 b_2013-03-14.csv (4bdbd288ee6f...)

Compute resources

Wall time 187.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 00:39Z.